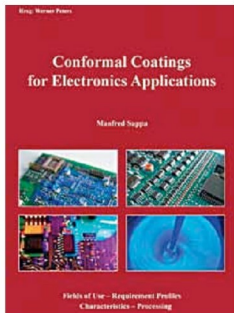


Pretreatment and Processing for Conformal Coating

Due to increasing complexities, precleaning and processing of boards before conformal coating becomes necessary to ensure the quality of the final product



DR MANFRED SUPPA

Precleaning of assembled printed circuit boards before application of the protective coating is, in most fields of the processing electronics industry, only carried out in exceptional cases since the no-clean soldering process has been introduced at the beginning of the 1990s. Thanks to the further developments of fluxes and solder pastes, a transition resistance of the solder residue of $>10^8$ could be maintained after the soldering process, even under high moisture loads. As a consequence, a precleaning process was no longer absolutely necessary and therefore often removed from the production process as a means of cost reduction.

In the course of the years, the complexity of printed circuit boards (components, low potential distances or multilayer) has developed to such an extent that the precleaning process

is being discussed and, in individual cases, introduced again as a process step prior to protective coating.

It is not easy to answer the question concerning the required cleanliness of the assembly. The intended use and the related climatic security are certainly central issues, just as the question of a possible coatability. Given its scheme logic, the assembly may require cleaning in respect of cross talk due to capacitive impurities or stray currents. Furthermore, not only the question concerning the quantity but also the type of particles to be removed has to be raised; it is a matter of ionic contamination, organic residues, and dusts and/or fibre residues.

The demands regarding cleanliness in accordance with the Joint Industry

Standard J-STD-001D are stated with a three-figure code beginning with the letter 'C,' followed by two figures. The first figure describes the specification for cleaning the surface (refer Table I). The second figure describes the test method to be used (see Table II).

The choice of a suitable cleaning process is almost as complex as the choice of the suitable coating process. For cleaning units and cleaning agents, an optimal tuning to the type of impurity is vital. There are three different cleaning systems available: organic

No-clean category

"Practically all of the modern low solid fluxes fall within the 'no-clean' category. They leave very little residue after soldering, meaning that there is no necessity for the soldered assemblies to be rinsed before a testing on bed-of-nails adaptors. For assemblies which require a high level of reliability or which are employed under critical conditions (high moisture/temperature), fluxes of the classes L0 or L1 in accordance with ANSI/JSTD-004 are used but as the complexity of PCBs are increasing exponentially precleaning process becomes necessary."

TABLE I
Specification for Cleaning the Surface

Cleanliness code	Test function
0	No side to be cleaned
1	Solder wave side to be cleaned
2	Both sides to be cleaned

TABLE II
Tests Regarding Surface Cleanliness in Accordance with the Joint Industry Standard J-STD-001D

Cleanliness code	Test function	Method
0	No test for cleanliness required	—
1	Test for resin residues required	IPC-TM-650, test method 2.3.27
2	Test for ionic residues required	IPC-TM-650, test method 2.3.25 or 2.3.25C
3	SIR test55 required	A documented method should be used
4	Test on other contaminations required	IPC-TM-650, test method 2.3.39
5	Tests other than those described	No specification